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COURSE MATERIAL

Interactive Diagram Demo: Breadth-First Search

Follow the queue across a small graph

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Queue trace

Breadth-first search from A

Green nodes are visited; red nodes are currently in the queue.

Enqueue the starting node

The queue contains A.

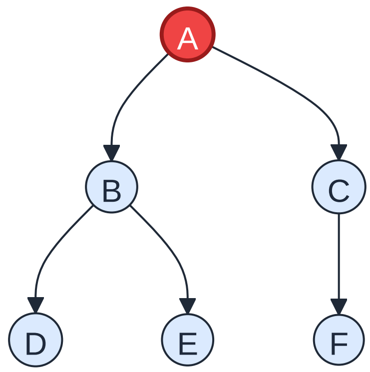

Figure 1: Enqueue the starting node

Visit A and enqueue its neighbors

A is visited; B and C enter the queue.

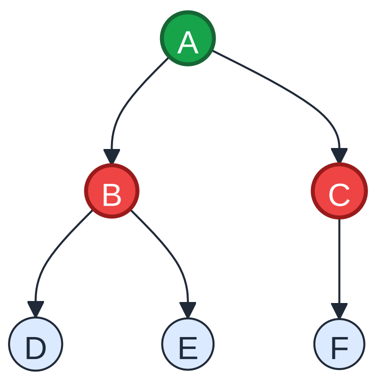

Figure 2: Visit A and enqueue its neighbors

Visit B and extend the queue

D and E are discovered behind C.

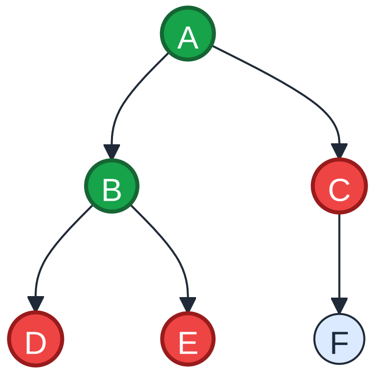

Figure 3: Visit B and extend the queue

Visit C and discover F

The remaining queue is D, E, F.

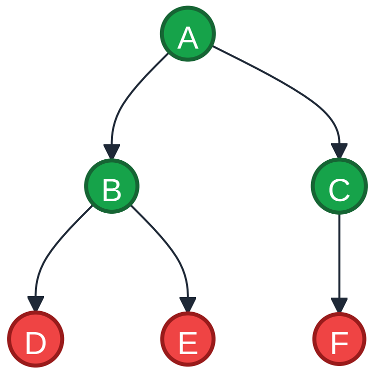

Figure 4: Visit C and discover F

Finish the traversal

Every reachable node has been visited in breadth-first order.

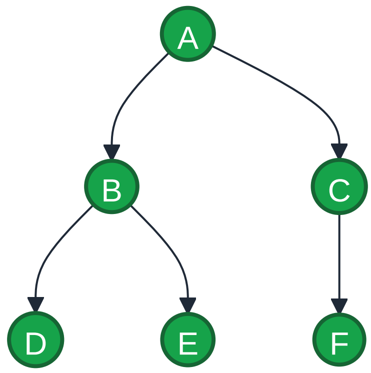

Figure 5: Finish the traversal

The step descriptions mirror the queue states `A` → `B, C` → `C, D, E` → `D, E, F` → empty.

The submission buttons above are placeholders. They open `example.com` and do not submit data.